Solutions to MIRA

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December 2025

Introduction

This document contains solutions to the textbook MIRA, available at measure.axler.net

Chapter 1

1A

1 We have $\sum_{j=1}^{n}(x_j - x_{j-1})\inf_{[x_{j-1},x_j]} f = \sum_{j=1}^{n}(x_j - x_{j-1})\sup_{[x_{j-1},x_j]} f$. Hence, f is constant on each interval. Since each interval intersects the next interval at the endpoints, f is constant.

2 $L(f, P, [a, b])$ is the sum of the lengths of intervals contained in (s, t). Clearly, the supremum of this is $t - s$. Similarly, $U(f, P, [a, b]$ is the sum of the lengths of intervals that intersect (s, t). Clearly, the infimum of this is $t - s$

3 See Theorem 6.6 of Rudin

4 See Theorem 6.12 of Rudin

5 See Theorem 6.12 of Rudin

6 We need to prove that a function that is zero except at finitely many points has zero integral. Let $|f(x)| \leq M$, with k nonzero points. We have $U(f, P_n, [a, b]) \leq 2kM/n$ and $L(f, P_n, [a, b] \geq -2kM/n$

7 RHS = sup $L(f, P_n, [a, b]) \leq$ LHS. Let P be a partition. We need to prove $\sup L(f, P_n, [a, b]) > L(f, P, [a, b]) - \epsilon$. So we need to show $L(f, P_n, [a, b]) > L(f, P, [a, b]) - \epsilon$ for some n.

Let P be $x_0, x_1, ..x_k$. Now consider

$$L(f, P_n, [a, b]) = \sum_{I} l(I)\ \inf_I f = \sum_{I \text{ containing } x_i} l(I)\ \inf_I f + \sum_{\text{other } I} l(I)\ \inf_I f$$

$$\geq -2(k+1)\frac{b-a}{2^n}M + \sum_{j=1}^{k} \inf_{[x_{j-1},x_j]} \sum_{I\subseteq(x_{j-1},x_j)} l(I)$$

For large enough n we have what we want.

8 Let $c_n = \frac{b-a}{n}\sum_{j=1}^{n} f(a+\frac{j(b-a)}{n})$. Let P_n be the partition into n equally spaced intervals. Then $L(P_n, f, [a,b]) \leq c_n \leq U(P_n, f, [a,b])$. By Problem 7 and the sandwich theorem, we have what we want.

9 Let $\epsilon > 0$. There is a partition P of $[a,b]$ with $U(f,P) - L(f,P) < \epsilon$, and we can assume this partition has the points c and d. Restricting this partition to $[c,d]$ does the job.

10 Problem 9 proves one direction of the required equivalence. To prove the other direction of the equivalence, let $\epsilon > 0$. We get partitions P_1 and P_2 of $[a,c]$ and $[c,b]$ with $U(P_1, f, [a,c]) - L(P_1, f, [a,c]), U(P_2, f, [c,b]) - L(P_2, f, [c,b]) < \epsilon/2$. Clubbing these partitions accomplishes the task. Now the integral over $[a,b]$ is the supremum of $L(P, f, [a,b])$ where P is a partition containing c. This can be split into 2 parts which is the RHS

11 See Rudin 6.20

12 $-|f| \leq f \leq |f|$, so $-\int|f| \leq \int f \leq \int|f|$

13 $U(f, P, [a,b]) - L(f, P, [a,b]) = \sum(f(x_j) - f(x_{j-1}))(x_j - x_{j-1})$. Now use the equally spaced partition P_n

14 See Rudin 7.16

1B

1 All $L(P,f)$ are zero. The number of rational numbers with denominator k is at most $k+1$. Hence, the number of rational numbers with denominator at most k is $\leq (k+1)(k+2)/2$. $U(P_n, f) = \frac{1}{n}\sum M_j \leq \frac{1}{n}((k+1)(k+2) + \frac{n}{k+1})$ which can be made arbitrarily small

2 $L(-f, P, [a,b]) = -\sum(x_j - x_{j-1})\sup_{[x_{j-1},x_j]} f = -U(P, f, [a,b])$

3 $L(f+g, P, [a,b] \geq L(f, P, [a,b]) + L(g, P, [a,b])$. Now, take the supremum.

4 Take $\chi_{\mathbb{Q}}$ and $\chi_{\mathbb{R}\setminus\mathbb{Q}}$

5 See example 7.6 of Rudin

Chapter 2

2A

1 LHS $\leq$ RHS by countable sub additivity. The other direction follows by inclusion

2 $I_1, I_2, ..$ are open intervals that cover A if and only if $tI_1, tI_2, ..$ are open intervals that cover tA.

3 $B = (B \setminus A) \sqcup (B \cap A)$. Now use countable sub additivity

4 See Heine-Borel theorem

5 See Theorem 26.9, Munkres

6 $(a, b) \cup \{a\} \cup \{b\} = [a, b]$. Now use countable sub additivity and inclusion.

7 The implication from LHS to RHS is obvious. For the other direction, see Problem 11

8 LHS $\leq$ RHS follows by countable sub additivity. Let $I_1, I_2..$ be open intervals that cover A. Then $I_1 \cap (-t, t), I_2 \cap (-t, t)...$, cover $A \cap (-t, t)$, $I_1 \cap (t, \infty), I_2 \cap (t, \infty)...$, cover $A \cap (t, \infty)$, $I_1 \cap (-\infty, -t), I_2 \cap (-\infty, -t)...$, cover $A \cap (-\infty, -t)$. We have $\sum l(I_k) \geq \sum l(I_k \cap (-t, t) + \sum l(I_k \cap (t, \infty) + \sum l(I_k \cap (-\infty, -t) \geq |A \cap (-t, t)| + |A \cap (t, \infty)| + |A \cap (-\infty, t)|$. Now use countable sub additivity

10 Use countable sub additivity and inclusion

11 Let $\{J_k\}$ be open intervals covering $\bigcup I_k$. We need to show

$$-\epsilon + \sum_{k=1}^{N} l(I_k) \leq \sum_{k=1}^{M} l(J_k)$$

for some M. Let $I_k = (a_k, b_k)$.

It suffices to show

$$\sum_{k=1}^{N} l(a_k + \frac{\epsilon}{2N}, b_k - \frac{\epsilon}{2N}) \leq \sum_{k=1}^{M} l(J_k)$$

$(a_k + \frac{\epsilon}{2N}, b_k - \frac{\epsilon}{2N})$ is covered by finitely many intervals $\{J_i\}_{i \in S_k}$

$(a_k + \frac{\epsilon}{2N}, b_k - \frac{\epsilon}{2N}) = \bigcup_{i \in S_k} J_i \cap (a_k + \frac{\epsilon}{2N}, b_k - \frac{\epsilon}{2N})$, so

$$\sum_{k=1}^{N} l(a_k + \frac{\epsilon}{2N}, b_k - \frac{\epsilon}{2N}) \leq \sum_{k=1}^{N} \sum_{i \in \cup S_k} l(J_i \cap (a_k + \frac{\epsilon}{2N}, b_k - \frac{\epsilon}{2N}))$$

Now $\{J_i \cap (a_k + \frac{\epsilon}{2N}, b_k - \frac{\epsilon}{2N})\}_k$ are disjoint open intervals within J_i. Hence we have $l(J_i) \geq \sum_k l(J_i \cap (a_k + \frac{\epsilon}{2N}, b_k - \frac{\epsilon}{2N}))$. Hence, switching the order of the summation we get what we want.

12 It is obvious that this set is closed. $(a,b) \subseteq \mathbb{R} \setminus \bigcup (r_k - \frac{1}{2^k}, r_k + \frac{1}{2^k})$ is not possible since (a,b) contains a rational. Countable subadditivity tells us $|F| = \infty$

13 Let $q_1, q_2, ..$ be the rationals in $[0,1]$. Let $G = \bigcup_{k=1}^{\infty} (q_k - \frac{\epsilon}{2^{k+1}}, q_k + \frac{\epsilon}{2^{k+1}})$. $|G| \leq \epsilon$. Then $[0,1] \setminus G$ is the required set

14 It is not a straight line

2B

1 This is obvious

2 Simple verification

3 We show that $\mathcal{S}$ contains (a,b). Let $a_1, a_2, ..$ be rationals strictly decreasing to a and $b_1, b_2, ..$ be rationals strictly increasing to b. Then $\bigcup (a_i, b_i]$=(a,b)

4 We need to show $\mathcal{S}$ contains (a,∞). Let $a_1, a_2, ..$ be rationals strictly decreasing to a. Let $n > a_1$ Then $\bigcup (a_i, n+i] = (a,\infty)$

5 $\mathcal{S}$ clearly contains (r,∞) for $r \in \mathbb{Q}$. Now taking strictly decreasing rationals gets (a,∞)

6 Strictly decreasing rationals again

7 $x \to x+t$ is measurable

8 $x \to tx$ is measurable

9 Consider the σ algebra on $\mathbb{R}$ given by $\{\phi, (-\infty,0), [0,\infty), \mathbb{R}\}$. Consider the function $f : \mathbb{R} \to \mathbb{R}$ which is 1 on $[-1,1]$ and -1 elsewhere. This function is not measurable, but $|f|$ is.

10 Let $A_k = \{0.x_1x_2x_3...|\ x_k = 5\}$. This is a Borel set. Then for $k_1 < k_2... < k_n$, $A_{k_1} \cap A_{k_2} ... \cap A_{k_n}$ is a Borel set. But this set is $\{0.x_1x_2...|x_{k_i} = 5\}$. Hence $\{0.x_1x_2...|x_i = 5 \text{ for at least n indices}\}$ is a Borel set. Now take the intersection for all n

11 This is trivial

12 This is also trivial

13 If $E_i \in \mathcal{S}$ then χ_{E_i} are measurable and hence any linear combination also. Suppose $\sum c_i \chi_{E_i}$ is measurable. This function is c_i on E_i and zero elsewhere. Hence $E_i \in \mathcal{S}$

14 This is trivial

15 It is easy to see that this is a σ algebra. Suppose $f : X \to \mathbb{R}$ is not constant on E_k, with $f(x) < f(y)$. Then $f^{-1}(f(x),\infty) \notin \mathcal{S}$

16 It is easy to see that this is a σ algebra. Let $f : X \to \mathbb{R}$. f is measurable with respect to $\mathcal{S}_A$ iff $f^{-1}(a,\infty) \in \mathcal{S}_A$ which happens iff f is $\mathcal{S}$-measurable and $f(A) \subseteq (a,\infty)$ or $f(A) \subseteq (-\infty, a]$ for all a. This gives us what we want.

17 We show that $f^{-1}(a,\infty)$ is Borel.

Suppose $f(x) > a$ and f is continuous at x. Then $f(x) > a$ for all $x \in (x - \delta_x, x + \delta_x) \cap X$.

$f^{-1}(a,\infty) = f^{-1}(a,\infty) \cap \{x : f \text{ is discontinuous at } x\} \cup f^{-1}(a,\infty) \cap \{x : f \text{ is continuous at } x\}$

$= f^{-1}(a,\infty) \cap \{x : f \text{ is discontinuous at } x\} \cup \bigcup (x - \delta_x, x + \delta_x) \cap X$ which is Borel

18 Define $f_n(x) = n(f(x + \frac{1}{n}) - f(x))$ and use 2.48

19

Claim. *The measurable functions are those that are constant on a co-countable set*

Proof. It is clear that any such function is measurable.

Suppose f is measurable. Then $f^{-1}(a,\infty)$ is countable for $a \in A$ and $f^{-1}(-\infty, a]$ is countable for $a \in B$. Let $c = \inf A$. Since $f^{-1}(c + \frac{1}{n}, \infty)$ is countable, $c \in A$ □

20 Use log

21 Let $\mathcal{T} = \{A \subseteq [-\infty,\infty] : f^{-1}A \in \mathcal{S}\}$. It is easy to see that $\mathcal{T}$ is a σ algebra. Now we have that $\mathcal{T}_{\mathbb{R}}$ contains every Borel set. We also have $\bigcap f^{-1}(n,\infty] = f^{-1}\infty \in \mathcal{S}$

22 Suppose f is not continuous at $x \in B$. Then it is not continuous from the right or from the left.

If it is not continuous from the left, then for all $\delta > 0, (x - \delta, x) \cap B \neq \phi$. Define the interval $I_x = (\sup_{y<x}\{f(y)\}, f(x))$

If it is not continuous from the right, then for all $\delta > 0, (x + \delta, x) \cap B \neq \phi$. Define the interval $I_x = (f(x), \sup_{y>x}\{f(y)\})$

It is not hard to see that the intervals I_x are disjoint. Hence there must be only countably many.

23 Let $f(x) \in f(\mathbb{R})$. Let $\epsilon > 0$. Define $\delta = f(x + \epsilon) - f(x)$. Suppose $f(z) \in (f(x), f(x) + \delta)$. Then $x < z < x + \epsilon$. This proves continuity from the right. Continuity from the left is analogous.

24 Use 23

25 Define $f_n(x) = f(x) + \frac{x}{n}$.

26 Define $g(x) = \begin{cases} \inf f & \text{if } \quad x < b \text{ for all b} \\ \sup_{\ y \leq x} f(y) & \text{otherwise} \end{cases}$

27 Let X be a set with at least 2 elements. Consider the σ algebra consisting of only X and ϕ, and let $x \in X$. Consider the function $f : X \to [-\infty, \infty]$ that sends x to ∞ and other elements to $-\infty$.

28 This is trivial

29 Define $g(x) = \frac{1}{2} + \frac{arctan(x)}{\pi}$.

Now define $f_t(x) = \begin{cases} g(x)\chi_{\mathbb{Q}}(x) \text{ if } t < 0 \\ g(x-t)\chi_{\mathbb{Q}}(x-t) \text{ otherwise} \end{cases}$

Now, consider the measurable space $\mathbb{R}$ with the countable-co-countable σ algebra

30 This is trivial

2C

1 $\mu(X)$ is the greatest element.

2 trivial

3 Define $w_i = \frac{1}{2^i}$ and $\mu(E) = \sum_{i \in E} w_i$

4 $(\mathbb{Z}, 2^{\mathbb{Z}}, \mu)$ where $\mu(E) = \sum_{i \in E} w_i$ and w_i is $\frac{1}{2^i}$ for $i \geq 1$ and 3 otherwise

5 Consider $\{A \mid \mu(A) > \frac{1}{n}\}$ which must be finite

6

Claim. *The only such c is $c = 4$*

Proof. Consider the measure space $(\mathbb{Z}_{\geq 0}, 2^{\mathbb{Z}_{\geq 0}}, \mu)$ where $\mu(E) = \sum_{i \in E} w_i$ and w_i is $\frac{1}{2^i}$ for $i \geq 1$ and $w_0 = 3$

Suppose $(X, \mathcal{S}, \mu)$ is such a measure space. $\mu(E_1) = 1$,so $\mu(X \setminus E_1) = c - 1 \geq 3$

Suppose $c > 4 + \epsilon$. Then $3 < c - 1 - \epsilon < c$, so $\mu(E_2) = c - 1 - \epsilon$. Then $\mu(X \setminus E_2) = 1 + \epsilon$

□

7 Consider the sequence $\frac{1}{2}, 3, \frac{1}{4}, 4, \frac{1}{8}, 5....$ The corresponding measure space is as required.

8 Consider the measurable space $(\mathbb{R}, \mathcal{S})$ where $\mathcal{S}$ is all unions of $(-\infty, 0], (0, \frac{1}{2}], (\frac{1}{2}, 1), [1, \frac{3}{2}), [\frac{3}{2}, \infty)$. Define the measures μ and ν by their values on these sets, respectively $1, 1, 1, 1, 1$ and $1, 2, 0, 2, 1$. Now define $\mathcal{A} = \{(0, 1), (\frac{1}{2}, \frac{3}{2})\}$

9 trivial

10 Consider the measure space from 2B.1 and $E_i = [i, \infty)$

11 trivial

2D

1 Define $A_n = \{0.x_1x_2... \ : \text{ there are 100 consecutive 4s in first n digits}\}$

Define $f(n)$ to be the number of n length decimal strings with 100 consecutive 4s. We have $f(n) = 10f(n-1) + 10^{n-100} - 10^{n-101} - 9f(n-101)$

Now $A_n = \bigcup_s \{0.sx_{101}x_{102}...\}$ where s ranges over all n length strings with 100 consecutive 4s. Each term in this union is a closed interval of length 10^{-n}. So $\mu(A_n) = f(n)10^{-n}$

We want $\mu(\bigcup A_n) = \lim \mu(A_n)$ since A_n are ascending.

Hence the answer is 1

2 Let B be a bounded non Lebesgue measurable set. There is $\epsilon > 0$ such that $|B \setminus F| \geq \epsilon$ for all closed $F \subseteq B$. Let $A = \frac{1}{\epsilon}B$. Let $F \subseteq A$ be closed. Suppose $|A| - |F| < 1$. Then $|B| - |\epsilon F| < \epsilon$, so $|B \setminus \epsilon F| < \epsilon$ which is a contradiction

5 Use 2.71c and take the union of the first n sets as F_n

6 Suppose A is Lebesgue measurable. Let $\epsilon > 0$. Then there is an open set $A \subseteq G_0$ with $|G_0 \setminus A| < \frac{\epsilon}{2}$. We know that $G_0 = I_1 \cup I_2 \cup I_3$.. where these are disjoint bounded open intervals. From 2A.11, $|G_0| = \sum l(I_i)$, so there is N such that $\sum_{i=1}^N l(I_i) > |G_0| - \frac{\epsilon}{2}$. Let $G = \bigcup_{i=1}^N I_i$. Now $|A \setminus G| + |G \setminus A| \leq |G_0 \setminus G| + |G_0 \setminus A| < \epsilon$

Suppose the condition holds. Let $\epsilon > 0$. Get G_0 such that $|A \setminus G_0| + |G_0 \setminus A| + \delta = \epsilon$. Get $I_1, I_2, ..$ such that I_i cover $A \setminus G_0$ and $\sum l(I_i) < |A \setminus G_0| + \delta$. Now let $G = G_0 \cup \{I_i\}$. We have $|G \setminus A| \leq |G_0 \setminus A| + \sum l(I_i) < \epsilon$

7 Use 2.71f and take the intersection of the first n sets as G_n

8 $A = B \cup (A \setminus B)$. So $t + A = (t + B) \cup (t + A \setminus B)$. $t + B$ is Borel, and outer measure is translation invariant.

9 $A = B \cup (A \setminus B)$. So $tA = tB \cup t(A \setminus B)$. tB is Borel and $t(A \setminus B)$ has outer measure 0.

10 $B = C \cup (B \setminus C)$, so $A \cup B = A \cup C \cup (B \setminus C)$. Hence $|A \cup B| = |A \cup C| = |A| + |C| = |A| + |B|$

14 $\frac{1}{4} = (0.\overline{02})_3$ and $\frac{9}{13} = (0.\overline{200})_3$

15 $13/17 = \frac{501\frac{16}{17}}{3^6} = (0.200120...)_3$

16 $G_4 = (\frac{1}{81}, \frac{2}{81}) \cup (\frac{7}{81}, \frac{8}{81}) \cup (\frac{19}{81}, \frac{20}{81}) \cup (\frac{25}{81}, \frac{26}{81}) \cup (\frac{55}{81}, \frac{56}{81}) \cup (\frac{61}{81}, \frac{62}{81}) \cup (\frac{73}{81}, \frac{74}{81}) \cup (\frac{79}{81}, \frac{80}{81})$

17 Every $(0.x_1x_2..)_3$ is the sum of the expansion with 2 converted into 1 and the expansion with 2 converted into 1 and 1 converted into 0.

20 $\Lambda(\frac{9}{13}) = (0.\overline{100})_2$, and $93/100 = (0.221002..)_3$, so $\Lambda(93/100) = (0.111)_2$

21 $\frac{1}{3} = (0.\overline{01})_2$, so $\Lambda^{-1}(\{\frac{1}{3}\}) = (0.\overline{02})_3$
$\frac{5}{16} = (0.0101)_2 = (0.0100\overline{1})_2$, so $\Lambda^{-1}(\{\frac{5}{16}\}) = [\frac{19}{81}, \frac{20}{81}]$

2E

1 Let $X = \{x_1, ...x_n\}$. Let $\epsilon > 0$. There is N_i such that $|f_k(x_i) - f(x_i)| < \epsilon$ for $k \geq N_i$. Then $|f_k(x) - f(x)| < \epsilon$ for $k \geq max\ N_i$

2 Define $f_n(m) = \frac{m}{n}$

3 Define $f_n(x) = \begin{cases} \frac{1}{x} \text{ if } x \geq \frac{1}{n} \\ n^2x \text{ otherwise} \end{cases}$

4 Let $\epsilon > 0$. There is N such that $|f_k(x) - f(x)| < \frac{\epsilon}{3}$ for $k \geq N$. Get $\delta > 0$ such that $|f_N(x) - f_N(y)| < \frac{\epsilon}{3}$ if $|x - y| < \delta$. Suppose $|x - y| < \delta$. Then $|f(x) - f(y)| \leq |f(x) - f_N(x)| + |f_N(x) - f_N(y)| + |f(y) - f_N(y)| < \epsilon$

6 Let $\epsilon > 0$. Define $A_{m,n} = \bigcap_{k=m}^{\infty} \{x : f_k(x) > n\} \in \mathcal{S}$. Now $A_{1,n} \subseteq A_{2,n}..$ and $\bigcup_m A_{m,n} = X$, so $\lim \mu(A_{m,n}) = \mu(X)$. So there is m_n such that $\mu(X) - \mu(A_{m_n,n}) < \frac{\epsilon}{2^n}$. Let $E = \bigcap_n A_{m_n,n}$. Then, as in the text, $\mu(X \setminus E) < \epsilon$.

Now we must verify uniform convergence. Let $n > 0$. Since $E \subseteq A_{m_n,n}$, $f_k(x) > n$ for all $k \geq m_n$ and $x \in E$

7

Claim (Dini)**.** *Let X be compact and let $g_1, g_2..$ be an increasing sequence of continuous functions on X that converges pointwise to g, which is continuous. Then the convergence is uniform.*

Proof. Let $\epsilon > 0$. Define $U_n = \{x : g(x) - g_n(x) < \epsilon\}$. Then these sets form an open cover, and they are also ascending. Since X is compact, $U_N = X$ □

8 It suffices to prove, by Problem 1, that there is a finite set E with $\mu(\mathbb{Z}_{>0} \setminus E) < \epsilon$, but this is just the tail sum.

9 Use Theorem 18.3 of Munkres

10 Suppose F is not closed. Then there is $x_0 \notin F$ which is a limit point of F. Define the function $f : F \to \mathbb{R}$ by $f(x) = \frac{1}{x - x_0}$.

11 Suppose F is not closed. Then there is $x_0 \notin F$ which is a limit point of F. Define the functions $f_n : F \to \mathbb{R}$ by $f_n(x) = sin\frac{1}{n(x-x_0)}$

Chapter 3

3A

1 Let P be the partition $X \setminus E, E$. Then $\infty > \mathcal{L}(f, P) \geq \infty \, \text{inf}_E f$, so the required result follows.

2 We have $\mathcal{L}(f, P) = \text{inf}_A f \leq f(c)$ where $A \in P$ is the set containing c. Define $A_n = f^{-1}(f(c) - \frac{1}{n}, f(c)] \in \mathcal{S}$. Now $\int f d\delta_c \geq \text{inf}_{A_n} f \geq f(c) - \frac{1}{n}$

3 The integral is positive if and only if $\mathcal{L}(f, P) > 0$ for some partition.

$\mu(f^{-1}(0, \infty)) > 0$ if and only if $\mu(f^{-1}(\frac{1}{n}, \infty)) > 0$ for some n

4 Consider the function $f : [0, 1] \to (0, 1]$

given by $f(x) = \begin{cases} \frac{1}{n}, \text{ where } n \text{ is smallest such that } x = \frac{m}{n} \text{ for some } m, \text{ if } x \in \mathbb{Q} \\ 1 \text{ otherwise} \end{cases}$

Then $L(f, P, [0, 1]) = 0$. This function is Borel measurable.

5 Let P be a partition of $\mathbb{Z}_{>0}$. Then $\mathcal{L}(b, P) = \sum |A_j| \, \text{inf}_{A_j} b = \sum_{j:A_j \text{ is finite}} |A_j| \, \text{inf}_{A_j} b \leq \sum b_i$

Also, $\sum_{i=1}^{n} b_i \leq \mathcal{L}(b, P)$ where P is the partition $\{1\}, ...\{n\}, \{n + 1, ...\}$

6 We have $\mathcal{L}(f, P) = \sum \mu(A_j) \text{inf}_{A_j} f = \sum \sum \mu(B_{i,j}) \text{inf}_{A_j} \leq \mathcal{L}(f, P')$

7 We have $\mathcal{L}(f, P) = \sum_{j=1}^{m} \mu(A_j) \text{inf}_{A_j} f = \sum_{j=1}^{m} \text{inf}_{A_j} f \sum_{x \in A_j} w(x) \leq \sum_{j=1}^{m} \sum_{x \in A_j} w(x) f(x)$

We now need to prove $-\epsilon + \sum_{j=1}^{m} \sum_{x \in A_j} w(x) f(x) < \sum_{x \in X_0} w(x) f(x)$ for some finite set X_0. But this is true.

Also $\sum_{i=1}^{n} w(x_i) f(x_i) \leq \mathcal{L}(f, P)$ where P is the partition $\{x_1\}, \{x_2\}, .., \{x_n\}, X \setminus \{x_1, ..x_n\}$

8 Let $f_k = \chi_{[k,k+1]}$

9 Let $E_1, E_2, .. \in \mathcal{S}$ be disjoint. Then $\nu(\bigcup E_i) = \int \chi_{\bigcup E_i} f d\mu = \int (\sum \chi_{E_i} f) d\mu$. By the monotone convergence theorem, this is $lim \int (\sum_{i=1}^{n} \chi_{E_i} f) d\mu = lim \sum_{i=1}^{n} \int \chi_{E_i} f d\mu = \sum v(E_i)$

10 Follows directly from the monotone convergence theorem

11 Let $g = \sum_{k=1}^{\infty} |f_k|$. By Problem 10, $\int g d\mu < \infty$. So $\mu(g^{-1}\infty) = 0$. Define $E = g^{-1}[0, \infty)$.

15 a) First, suppose f is nonnegative. Now $\{\mathcal{L}(f,P)\} = \{\mathcal{L}(f^+,P)\}$, so $\int f_t d\lambda = \int f d\lambda$

Now let f be general. $\int f_t d\lambda = \int f_t^+ d\lambda - \int f_t^- d\lambda = \int f d\lambda$

b) First, suppose f is nonnegative. Now $\mathcal{L}(f_t,P) = \frac{1}{|t|}\mathcal{L}(f,P_t)$, so $\int f_t d\lambda = \frac{1}{|t|}\int f d\lambda$

Now let f be general. $\int f_t d\lambda = \int f_t^+ d\lambda - \int f_t^- d\lambda = \frac{1}{|t|}\int f d\lambda$

16 Since any $\mathcal{S}$-partition is also a $\mathcal{T}$-partition, LHS $\leq$ RHS.

Now, let $P = E_1, E_2, ..E_n$ be a $\mathcal{T}$-partition, with $\mu_2(E_i) > 0$ for $i = 1,..,t$

Let $c_1 < c_2 < .. < c_k$ be the distinct values of $\inf_{E_i} f = c_{m_i}$ for $i = 1..t$. Now $f^{-1}[c_1,c_2), ...f^{-1}[c_{k-1},c_k), f^{-1}[c_k,\infty] \in \mathcal{S}$ are disjoint.

We can get an $\mathcal{S}$-partition $P' = S_1, S_2, ..S_k, S_{k+1}$ by adding another set (note that this set will have measure 0).

Now $\mathcal{L}(f,P) = \sum_{i=1}^t \mu_2(E_i)\inf_{E_i} f = \sum_{i=1}^t c_{m_i} \sum_{j=m_i}^k \mu_2(S_j \cap E_i)$

We also have $\mathcal{L}(f,P') = \sum_{j=1}^k \mu_2(S_j)\inf_{S_j} f = \sum_{i=1}^t \sum_{j=m_i}^k \mu_2(S_j \cap E_i)\inf_{S_j} f$

19 $\inf f \leq f \leq \sup f$, so we can use 3.8 to get the required result.

3B

1 Let $f_k = e_k$

2 Define $f_k(x) = \begin{cases} 0 \text{ if } x < k \\ x-k \text{ if } x \in [k,k+1] \\ 1 \text{ otherwise} \end{cases}$

3 Let $\epsilon > 0$. By 3.28, there is $\delta > 0$ such that $\int_B |f| d\lambda < \epsilon$ for every Borel set with $\lambda(B) < \delta$. Let $y < x$. Then

$$g(x) - g(y) = \int \chi_{(-\infty,x)} f d\lambda - \int \chi_{(-\infty,y)} f d\lambda = \int \chi_{[y,x)} f d\lambda = \int_{[y,x)} f d\lambda$$

Hence $|g(x) - g(y)| \leq \int_{[y,x)} |f| d\lambda$

5 Let $\epsilon > 0$. By 3.29 there is a Borel set E with $|E| < \infty$ and $\int_{\mathbb{R}\setminus E} |f| d\lambda < \frac{\epsilon}{3}$.

By 3.28 there is $\delta > 0$ such that $\int_B |f| d\lambda < \frac{\epsilon}{3}$ for every Borel set B with $|B| < \delta$

Now $E \cap [-k,k]$ is an increasing sequence of sets with union E. Thus by 2.59 there is N such that for $k \geq N, |E \cap [-k,k]| > |E| - \delta$

Now $|\int f d\lambda - \int_{[-k,k]} f d\lambda| = |\int_E f d\lambda + \int_{\mathbb{R}\setminus E} f d\lambda - \int_{[-k,k]} f d\lambda|$

We have $|\int (\chi_E - \chi_{[-k,k]}) f d\lambda| \leq \int \chi_{E\setminus[-k,k]} |f| d\lambda + \int \chi_{[-k,k]\setminus E} |f| d\lambda < \frac{2\epsilon}{3}$ for $k \geq N$. Now use the triangle inequality.

9 Use 2B.13 and 3.15

11 The given set is $|f|^{-1}(0,\infty) = \bigcup_{n=1}^{\infty} |f|^{-1}(\frac{1}{n},\infty)$. Call this set A_n

Now, $\int |f| d\mu \geq \mathcal{L}(\{A_n, X \setminus A_n\}, |f|) \geq \frac{\mu(A_n)}{n}$

Chapter 4

4A

1 We may assume $|h|^p \in \mathcal{L}^1(\mu)$. By Markov's inequality, we have $\mu(\{x \in X : |h(x)|^p \geq c^p\}) \leq \frac{1}{c^p}\int |h|^p$. We get what we want since $x \to x^p$ is an increasing function

2 Let $a = \int h d\mu$. Then by Problem 1, LHS $\leq \frac{1}{c^2}\int (h-a)^2 d\mu$

Now, $\int (h^2 + a^2 - 2ah) d\mu = \int h^2 d\mu - a^2$, since $\mu(X) = 1$

3 Suppose c satisfies the condition. Let $E_c = \{x : |h(x)| \geq c\}\}$. Then $\int |h| - c\chi_{E_c} = 0$. This means, by 3A.3 that $\mu\{x : |h(x)| > c\chi_{E_c}(x)\} = 0$. Let $E = \{x : |h(x)| = c\}$. Then $\mu(X \setminus E) = 0$. So there cannot be more than one such c

4 Let $\delta > 0$. We show that $3-\delta$ does not work. Consider the intervals $(0,1)$ and $(1-\epsilon, 2)$. Now $(3-\delta)*(0,1) = (-1+\frac{\delta}{2}, 2-\frac{\delta}{2})$ and $(3-\delta)*(1-\epsilon,2) = (\frac{\delta}{2} - \epsilon(2-\frac{\delta}{2}), 3+\epsilon-\frac{\delta(1+\epsilon)}{2})$

5

Claim. *Suppose I_1 and I_2 are nonempty bounded open intervals that are not disjoint and $|I_1| \geq |I_2|$. Then $I_2 \subseteq 3 * I_1$*

Proof. Without loss of generality, assume $I_1 = (-\frac{L}{2}, \frac{L}{2})$ and $I_2 = (a,b)$ with $a > -\frac{L}{2}$. Then $b \leq L + a$. Now $3 * I_1 = (-\frac{3L}{2}, \frac{3L}{2})$. Since $a < \frac{L}{2}$, we have what we want. □

6 Suppose $b \in (0,1)$. Then for small t the value of the inner expression is 1, and this is the supremum.

Suppose $b \geq 1$. Then the inner expression is $\frac{1}{2t}\int_{b-t}^{b} \chi_{[0,1]}$. We consider only t such that $0 \leq b-t < 1$. Then the expression simplifies to $\frac{1-b+t}{2t} = \frac{1}{2} - \frac{b-1}{2t}$. This is maximized by selecting $t = b$.

Suppose $b \leq 0$. Then the inner expression is $\frac{1}{2t}\int_{b}^{b+t} \chi_{[0,1]}$. We consider only t such that $0 < b+t \leq 1$. Then the expression simplifies to $\frac{1}{2} - \frac{-b}{2t}$. This is maximized by selecting $t = 1-b$

7 Clearly on $(0,1)\cup(2,3)$ the value of the function is 1.

Suppose $b \le 0$. Then the inner expression is $\frac{1}{2t}\int_b^{b+t}\chi_{[0,1]\cup[2,3]}$. We consider t such that $0 < b+t \le 1$ or $2 < b+t \le 3$. In the first case the expression simplifies to $\frac{1}{2} - \frac{-b}{2t}$. This is maximized by selecting $t = 1-b$, and we get $\frac{1}{2(1-b)}$. In the second case the expression simplifies to $\frac{1}{2} - \frac{1-b}{2t}$, which is maximized by selecting $t = 3-b$, for which we get $\frac{1}{3-b}$.

Suppose $b \in [1, \frac{3}{2}]$. For the inner expression, we consider only t such that $b - t < 1$. Then the inner expression is $\frac{1}{2t}(1 - b + t + \int_b^{b+t}\chi_{[2,3]})$.

Now if $2 < b+t \le 3$ this is $1 - \frac{1}{2t}$, so we choose $t = 3-b$ to get $1 - \frac{1}{2(3-b)} \ge \frac{1}{2}$

Otherwise, we have $\frac{1}{2} - \frac{b-1}{2t} \le \frac{1}{2}$, so we reject this.

The other cases follow by symmetry.

We have $$(\chi_{[0,1]\cup[2,3]})^*(b) = \begin{cases} \frac{1}{3-b} & b < -1 \\ \frac{1}{2(1-b)} & b \in [-1,0] \\ 1 & b \in (0,1) \\ 1 - \frac{1}{2(3-b)} & b \in [1, \frac{3}{2}] \\ 1 - \frac{1}{2b} & b \in (\frac{3}{2}, 2] \\ 1 & b \in (2,3) \\ \frac{1}{2(b-2)} & b \in [3,4] \\ \frac{1}{b} & b > 4 \end{cases}$$

8 Suppose $b \le 0$. Then the inner expression is $\frac{1}{2t}\int_0^{b+t} h$ and we consider only t such that $0 < b+t \le 1$. But this is $\frac{(b+t)^2}{4t}$. So we get $\frac{1}{4(1-b)}$

Suppose $b \ge 1$. Then the inner expression is $\frac{1}{2t}\int_{b-t}^{1} h$ and we consider only t such that $0 \le b-t < 1$. But this is $\frac{1}{4t}(1-(b-t)^2)$. So we get $\frac{b-\sqrt{b^2-1}}{2}$

Suppose $b \in (0, \frac{1}{2}]$. We consider only t such that $b+t \le 1$. If $b - t \ge 0$ the inner expression is b. If $b - t < 0$ the inner expression is $\frac{b}{2} + \frac{t}{4} + \frac{b^2}{4t}$. So we get $\frac{1}{4(1-b)}$

Suppose $b \in (\frac{1}{2}, 1)$. We consider only t such that $0 \le b - t$. If $b + t \le 1$ the inner expression is b. If $b+t > 1$ the inner expression is $\frac{b}{2} - \frac{t}{4} + \frac{1-b^2}{4t}$. So we get b.

Chapter 5

5A

1 Let π_1, π_2 be the projection functions. Note that $\{C \subseteq X \times Y : \pi_1(C) \in \mathcal{S}\}$ is a σ-algebra on $X \times Y$ and it contains $\mathcal{S} \otimes \mathcal{T}$. This proves $A \in \mathcal{S}$. Similarly, $B \in \mathcal{T}$

2 Note that $\{C \subseteq X \times X : \pi_1(C \cap Diag) \in \mathcal{S}\}$ is a σ-algebra containing $\mathcal{S} \otimes \mathcal{S}$

3 Let W be a non-Borel set. Define $E = \{(w, w) : w \in W\}$ and use Problem 2

4 The function $(x, y) \to xy$ is the product of the projection functions and is hence measurable. Now $h = Prod \circ (f, g)$. So $h^{-1}(a, \infty) = (f, g)^{-1}(Prod^{-1}(a, \infty))$.

Let A, B be Borel sets. Now $(f, g)^{-1}(A \times B) = f^{-1}(A) \times g^{-1}(B) \in \mathcal{S} \otimes \mathcal{T}$

5 The union of two non-disjoint intervals is an interval. So we may assume all intervals are pairwise disjoint. We have finite intervals $I_1, I_2, .., I_n$ with endpoints $a_1 \leq b_1 \leq a_2 \leq b_2 ... a_n \leq b_n$. In addition we have at most two infinite intervals.

6 Let $q_1, q_2, ..$ be the rationals. Consider the set $\bigcup (q_i - \frac{1}{2^i}, q_i + \frac{1}{2^i})$. This set has finite Lebesgue measure. Suppose its complement was a countable union of intervals. Then its complement is countable and has Lebesgue measure zero, which is a contradiction.

7 trivial

8 Suppose a) holds. Then we have Y_k with $X = \bigcup Y_k$ and $\mu(Y_k) < \infty$. Set $X_k = \bigcup_{i=1}^{k} Y_i$

Suppose b) holds. Then $X_1, X_2 \setminus X_1, X_3 \setminus X_2, ..$ make c) hold

c) implies a) by definition

9

5B

-

Chapter 6

6A

2 See Rudin 2.20

3 Metric is continuous

4 Open sets form a topology

5 Use de Morgan and Problem 4

6 a) follows by the definition of closure. For b) take any discrete metric space with more than one element.

7 Metrizable topologies are Hausdorff

9 Let U be open. Let $F_n = \{x : d(x, U^c) \geq \frac{1}{n}\}$.

10 The statement is true, since RHS is the smallest closed set containing U and W. Any such set must contain $\overline{U}$ and $\overline{W}$. But $\overline{U} \cup \overline{W}$ is closed.

14 Let $x_1, x_2, ..$ be the sequence. We have that $x_{n_1}, x_{n_2}, ... \to x$. Let $\epsilon > 0$. There is N such that $d(x_{n_k}, x) < \frac{\epsilon}{2}$ for $k \geq N$. There is M such that $d(x_n, x_m) < \frac{\epsilon}{2}$ if $n, m \geq M$. Choose $K \geq N$ so that $n_K \geq M$. Now let $n \geq n_K$. We have $d(x_n, x) \leq d(x_n, x_{n_K}) + d(x_{n_K}, x) < \epsilon$

6B

1 Multiply by complex conjugate

2 Let $z = a + ib$ and square the inequality, we get $\max\{a^2, b^2\} \leq a^2 + b^2 \leq 2\max\{a^2, b^2\}$

3 As in Problem 2, we get $a^2 + b^2 + 2|ab| \leq 2(a^2 + b^2) \leq 2(a^2 + b^2 + 2|ab|)$

9 trivial

10 trivial

6C

1 The norm is 1-Lipschitz by the reverse triangle inequality.

2 LHS $\subseteq$ RHS is 6A.6. Suppose $||f - g|| = r$, and let $\epsilon > 0$. Let $h = (1 - \frac{\epsilon}{2r})(g - f) + f$. Then $h \in B(f, r)$ and $||h - g|| = \frac{\epsilon}{2}$

3 We have $||(2, 2, ..)|| = n\sqrt{2} \neq 2||(1, 1, ..)|| = 2n$

We have $||e_i||_{1/2} = 1$, and $||e_1 + e_2||_{1/2} = 4$

4 Let $x_1, x_2, ..$ be Cauchy. Then there is N such that $d(x_n, x_N) < 1$ for $n \geq N$

5 We have $||v||_\infty \leq ||v||_1$, so if a sequence is Cauchy with respect to norm 1 it is Cauchy with respect to norm infinity. We also have $||v||_1 \leq n||v||_\infty$, so the converse also holds.

Suppose $v_1, v_2, ..$ is Cauchy. Then $v_1^i, v_2^i, ..$ is Cauchy and converges to v^i. Then $v_1, v_2..$ converges to v

6 It is clear that this is a valid norm. Suppose $f_1, f_2, ..$ is Cauchy. Then for each x, $f_1(x), f_2(x), ..$ is Cauchy. So we get pointwise convergence to a function f. It is clear that f is bounded. Now Theorem 7.8 of Rudin completes the proof

7 Consider the Cauchy sequence $(0, 0, ..), (\frac{1}{2}, \frac{1}{2}, 0, 0, ..), (\frac{2}{3}, \frac{2}{3}, \frac{2}{3}, 0, 0, ..)$. If it converged, it would converge to $(1, 1, ..)$ which is not in ℓ^1

8 Suppose $x_1, x_2, ..$ is Cauchy. Then $x_1^i, x_2^i, ..$ is Cauchy and converges to x^i. Now $\sum_{i=1}^n |x^i| = \lim \sum_{i=1}^n |x_k^i|$. But $\sum_{i=1}^n |x_k^i| \leq ||x_k||_1 \leq M$, by Problem 4. So $x \in \ell^1$.

Let $\epsilon > 0$. We have $\sum_{i=1}^n |x_k^i - x_m^i| \leq \epsilon$ if $k, m \geq N$. If we take the limit with m, we get $\sum_{i=1}^n |x_k^i - x^i| \leq \epsilon$ if $k \geq N$. So $||x_k - x||_1 \leq \epsilon$ for $k \geq N$

9 Define, for $n \geq 3$, $\quad f_n = \begin{cases} 0 & \text{for } x \leq \frac{1}{2} \\ n(x - \frac{1}{2}) & \text{for } x \in (\frac{1}{2}, \frac{1}{2} + \frac{1}{n}] \\ 1 & \text{for } x > \frac{1}{2} + \frac{1}{n} \end{cases}$

We see that f_n is a Cauchy sequence. Suppose it converges to f. This means $\int_0^1 |f - f_n| \to 0$. Hence we have $\int_0^{\frac{1}{2}} |f| = 0$ and since f is continuous, f is identically 0 on $[0, \frac{1}{2}]$

Similarly, we have $\int_{\frac{1}{2}+\frac{1}{n}}^1 |f - 1| \to 0$, so $\int_{\frac{1}{2}}^1 |f - 1| = 0$. This means f is identically 1 on $[\frac{1}{2}, 1]$, which is a contradiction since f is continuous.

10 We have $B(w, r) \subseteq U$. Let $v \in V$. We see that $\frac{r(v-w)}{2||v-w||} + w \in U$. So $v - w \in U$ and hence $v \in U$

11 A normed vector space is path connected, because we can take the line segment.

12 Let $f, g \in \overline{U}$. Then there are $f_1, f_2, .. \in U$ with $f_i \to f$ and $g_1, g_2, .. \in U$ with $g_i \to g$. Then $f_1 + g_1, f_2 + g_2, .. \in U$ and they converge to $f + g$. Similarly $cf \in \overline{U}$

6D

1 We have $\phi(v) = \alpha$. Then $\phi^{-1}(\alpha) = v + \text{null } \phi$. Now the required result follows from 6.52

2 Suppose $\phi(u) \neq 0$. Let $v \in V$. Now $v - \frac{\phi(v)}{\phi(u)} u \in \text{null } \phi$

3 One direction is obvious, and we may assume $\text{null}\phi = \text{null}\psi \neq V$. So we have $\psi(v) = \alpha\phi(v) \neq 0$. Then for any w, $v - \frac{\psi(v)}{\psi(w)} w \in \text{null } \psi$. Hence $\psi(w) = \alpha\phi(w)$

4 We have $||T(c_1, c_2, .., c_n)|| \leq \sum |c_i| ||Te_i|| \leq ||(c_1, c_2, .., c_n)||_\infty \sum ||Te_i||$

6E

1 Follows from Theorem 17.5 of Munkres

2 We have $\text{int } U = \overline{U^c}^c$

3 No, take $(0, 1]$ and $[1, 2)$

4 Trivially true

5 X is a closed subspace of a complete metric space. Baire's theorem is not violated because $\{\frac{1}{k}\}$ is open in X.

6 $\mathbb{Q}$

Chapter 7

7A

1 We have $||\alpha f||_\infty = \inf\{t > 0 : \mu(\{x \in X : |f(x)| > \frac{t}{|\alpha|}\}) = 0\} = |\alpha|\inf\{t > 0 : \mu(\{x \in X : |f(x)| > t\}) = 0\} = |\alpha|||f||_\infty$

Let $\epsilon > 0$. Then there is $t_1 < ||f||_\infty + \frac{\epsilon}{2}$, $t_2 < ||g||_\infty + \frac{\epsilon}{2}$ with $\mu(\{x : |f(x)| > t_1\}) = 0$ and $\mu(\{x : |g(x)| > t_2\}) = 0$

Now $\{x : |f(x) + g(x)| > t_1 + t_2\} \subseteq \{x : |f(x)| + |g(x)| > t_1 + t_2\} \subseteq \{x : |f(x)| > t_1\} \cup \{x : |g(x)| > t_2\}$. So $||f+g||_\infty \leq t_1 + t_2 < ||f||_\infty + ||g||_\infty + \epsilon$

2 As shown in the text, equality holds precisely when $a = b^{\frac{1}{p-1}}$, which is $a^p = b^{p'}$

3 Let the measure space be $\{1, .., n\}$ and $f(i) = a_i, h(i) = \frac{1}{n^{4/5}}$, with $p = 5$. Then applying Holder's inequality gives the required result.

4 We have $\mu\{x : |h(x)| > ||h||_\infty\} = 0$. So $|f|(||h||_\infty - |h|) \geq 0$ almost everywhere. If we integrate, we obtain the required result.

5 Clearly, we may assume $||f||_p, ||h||_{p'} \neq 0$. We have $||\tilde{f}\tilde{h}||_1 = \int \frac{|\tilde{f}|^p}{p} + \frac{|\tilde{h}|^{p'}}{p'}$ where $\tilde{f} = \frac{f}{||f||_p}$ and $\tilde{h} = \frac{h}{||h||_{p'}}$

This is equivalent to $\int \frac{|\tilde{f}|^p}{p} + \frac{|\tilde{h}|^{p'}}{p'} - |\tilde{f}\tilde{h}| = 0$. By Young's inequality, 3A.3 and Problem 2 this is equivalent to $|\tilde{f}(x)|^p = |\tilde{h}(x)|^{p'}$ almost everywhere

Conversely, suppose $|f(x)|^p = c|h(x)|^{p'}$ almost everywhere. Then it is easy to see that Holder's inequality is an equality.

6 We have

$$\int |fh| = \int |f|||h||_\infty \iff \int |f|(||h||_\infty - |h|) = 0$$

This integral is $\int \chi_E |f|(||h||_\infty - |h|)$ where $E = \{x : f(x) \neq 0 \text{ and } |h(x)| < ||h||_\infty\}$. Hence by 3A.3 this integral is zero if and only if $\mu(E) = 0$.

The given condition is $\mu\{x : f(x) \neq 0, |h(x)| \neq ||h||_\infty\} = 0$. This is equivalent by Problem 4.

7 Let $\tilde{f} = |f|^r$ and $\tilde{h} = |h|^r$. Then apply Holder's inequality

8

Claim. *Suppose* $\frac{1}{p_1}+..+\frac{1}{p_n}=\frac{1}{p}$. *Then* $||f_1f_2..f_n||_p \le ||f_1||_{p_1}..||f_n||_{p_n}$

Proof. Let $\frac{1}{p_1}+..+\frac{1}{p_n}=\frac{1}{q}$.
We have $||f_1f_2..f_{n+1}||_p \le ||f_{n+1}||_{p_{n+1}}||f_1f_2..f_n||_q$. Now apply the induction hypothesis. ☐

10 We need only prove b), and the case $q=\infty$ is trivial. It suffices to prove the following claim

Claim. *Let* $x_1,x_2,..,x_n>0$. *Define, for* $\alpha>0$, $g(\alpha)=(x_1^\alpha+..+x_n^\alpha)^{1/\alpha}$. *Then* g *is decreasing*

Proof. Let $\alpha_1<\alpha_2$. Define $p_i=\frac{x_i^{\alpha_1}}{g(\alpha_1)^{\alpha_1}}$. Then $\sum p_i=1$, so $\sum p_i^{\frac{\alpha_2}{\alpha_1}}\le 1$. If we expand this we get $g(\alpha_2)^{\alpha_2}\le g(\alpha_1)^{\alpha_2}$, so we are done. ☐

11 The harmonic sequence

7B

1 We have $||e_i||=1$ and $||e_1+e_2||=2^{\frac{1}{p}}>2$ so the triangle inequality is not satisfied.

2 Consider $\{(a_1,a_2,..,a_n,0,0,..) : a_i \text{ has rational coordinates}\}$ to see that ℓ^p is separable.

Consider $\{0,1\}^{\mathbf{Z}_+}$ to see that any dense subset of ℓ^∞ must be uncountable.

3 Let $S\subseteq \mathcal{L}^p(\mathbb{R})$ be step functions such that coefficients are in $\mathbb{Q}[i]$ and intervals have both endpoints rational.

Claim. $\tilde{S}$ *is dense in* $L^p(\mathbb{R})$

Proof. Let $\tilde{f}\in L^p(\mathbb{R})$ and $\epsilon>0$. By 7A.18 there is a step function $g\in \mathcal{L}^p(\mathbb{R})$ such that $||f-g||_p<\frac{\epsilon}{2}$. We put $g=a_1\chi_{I_1}+..+a_n\chi_{I_n}$ where we assume all intervals are disjoint and bounded. Now get nontrivial intervals $J_i\supset I_i$ with rational endpoints such that $\ell(J_i)-\ell(I_i)<(\frac{\epsilon}{4n|a_i|})^p$. Then get $b_1,b_2,..,b_n$ with rational coordinates such that $|a_i-b_i|<\frac{\epsilon}{4n(\ell(J_i))^{\frac{1}{p}}}$. Now let $h=b_1\chi_{J_1}+..+b_n\chi_{J_n}$. Now $||\tilde{f}-\tilde{h}||_p=||f-g+g-h||_p<\frac{\epsilon}{2}+\sum||a_i\chi_{I_i}-b_i\chi_{J_i}||_p\le\epsilon$ ☐

Consider $\{f:\mathbb{R}\to\{0,1\} : |f[n,n+1)|=1\}\subseteq\mathcal{L}^\infty(\mathbb{R})$. The image in $L^\infty(\mathbb{R})$ is uncountable and pairwise distance is 1.

Chapter 8

8A

1 To see definiteness, note that the only function identically zero on the rationals is the zero function.

3 If we expand the equation, we get $\text{Re}\langle f, g\rangle = 0$. To see that it doesn't work in the complex case, set $f = e^{\frac{i\pi}{4}}$ and $g = e^{-\frac{i\pi}{4}}$

4 Set $a = \frac{1}{2}(1, 6, 3)$ and $b = (4.5, 1, -3.5)$

5 Apply Cauchy-Schwarz to $(\frac{1}{\sqrt{a}}, \frac{1}{\sqrt{b}}, \frac{1}{\sqrt{c}}, \frac{1}{\sqrt{d}})$ and $(\sqrt{a}, \sqrt{b}, \sqrt{c}, \sqrt{d})$

9 arccos is defined only on $[-1, 1]$

11 We have $\langle sf+tg, sf+tg\rangle = \langle tf+sg, tf+sg\rangle \iff (s^2-t^2)(||f||^2-||g||^2) = 0$

12 $\langle f - g, f - g\rangle = 0$

13 trivial

14 trivial

8B

1 Define $s_n = (1, \frac{1}{2}, .., \frac{1}{n}, 0, 0, ..) \in \ell^1$. Then $s_1, s_2, ..$ is a Cauchy sequence. If it converged it would converge to $(1, \frac{1}{2}, ..) \notin \ell^1$

To see that $C[0, 1]$ is not a Hilbert space, the functions of 6C.9 suffice, with the same argument.

5 See page 31 of Rudin

8 Let $x, y \in \overline{U}$. Let $\epsilon > 0$. Get $x', y' \in U$ with $||x - x'||, ||y - y'|| < \epsilon$. Then $||tx + (1 - t)y - (tx' + (1 - t)y')|| \leq \epsilon$

9 Let $x, y \in \text{int } U$. Then $B(r, x), B(r, y) \subseteq U$. Let $z \in B(r, tx + (1 - t)y)$. Then $z - tx + (1 - t)y + x, z - tx + (1 - t)y + y \in U$. Now use convexity.